

COMP 1324: Algorithmnics

Linear Programming

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Optimization problem

- A company produces three products, x_1 , x_2 , and x_3 , with profit per unit equal to 3, 1, and 2 respectively.
- To produce x_1, x_2, x_3 , the company needs raw material, labour and machine time.
- Each unit of product x_1 requires 1 hours of labour, 2 hours of machine time, and 4 units of raw materials.
- Each unit of product x_2 requires 1 hour of labour, 2 hours of machine time, and 1 units of raw materials.
- Each unit of product x_3 requires 3 hour of labour, 5 hours of machine time, and 2 units of raw materials.
- The company has a total of 30 hours of labour, 24 hours of machine time, and 36 units of raw material available.
- How many units of each product should be produced to maximise profit? (fractions are allowed)

- Let z be the total profit then the optimisation problem can be written as

$$\begin{aligned} &\text{maximise } z = 3x_1 + x_2 + 2x_3 \\ &\text{subject to } x_1 + x_2 + 3x_3 \leq 30 \\ &\quad 2x_1 + 2x_2 + 5x_3 \leq 24 \\ &\quad 4x_1 + x_2 + 2x_3 \leq 36 \\ &\quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

- Our goal is to find the values of x_1, x_2, x_3 that maximise z subject to the constraints.
- This mathematical model is an example of a linear programming problem.

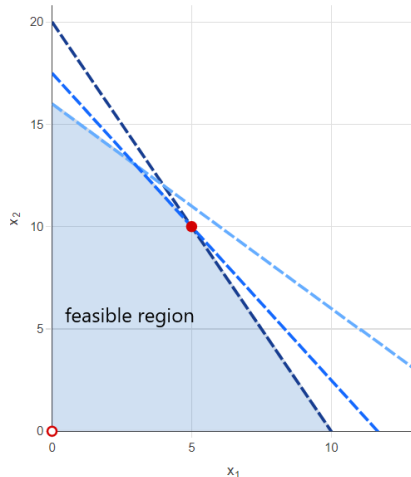
- Given a set of real numbers $a_1, \dots, a_n$ and a set of variables $x_1, \dots, x_n$, a linear function f on those variables is define by

$$f = a_1x_1 + \dots + a_nx_n = \sum_{i=1}^n a_ix_i$$

- Given a real number b and a linear function f as above, the inequalities $f \leq b$ and $f \geq b$ are called a **linear** inequalities.
- And $f = b$ is called a **linear** equality.
- A linear programming problem is the problem of maximising or minimising a **linear** function subject to a finite number of **linear** inequalities and/or equalities (constraints).

Feasible region

- In linear programming the constraints form a region in multidimensional space. The (2-d) region below is example of the **feasible region**.



Methods for solving LP

- There are many methods for solving LP problems
- The most known/used ones are
 - 1 Simplex method: finds a solution by moving along the edges of the feasible region.
 - 2 Interior point method: unlike simplex, IPM moves through the interior of the feasible region.
- We will restrict ourselves to simplex.

Standard and slack forms

- It is often convenient to write a linear programming problem in canonical forms
- A linear programming problem in **standard** form is the maximisation of a linear function subject to **linear inequalities**.
- A linear programming problem in **slack** form is the maximisation of a linear function subject to **linear equalities**.
- Note that we follow the terminology of your textbook **cormen2022introduction**. Other sources may use different terminology.

Standard form

- A linear programming problem in **standard** form is written as

$$\begin{aligned} &\text{maximise } z = \sum_{j=1}^n c_j x_j \\ &\text{subject to } \sum_{j=1}^n a_{ij} x_j \leq b_i && \text{for } i = 1 \dots m \\ & && x_j \geq 0 && \text{for } j = 1 \dots n \end{aligned}$$

- using matrix/vector notations $A = (a_{ij}), x = (x_j), b = (b_i)$, and $c = (c_j)$, the problem can be written as

$$\begin{aligned} &\text{maximise } z = c^T x \\ &\text{subject to } Ax \leq b \\ & && x \geq 0 \end{aligned}$$

Conversion to standard form

What if the LP problem is not in standard form? We convert it to standard form by using the following rules:

- For f objective function, $\min f$ is replaced by $\max -f$
- For any constraint, $\sum_{j=1}^n a_{ij}x_j = b$ is replaced with $\sum_{j=1}^n a_{ij}x_j \leq b$ and $\sum_{j=1}^n a_{ij}x_j \geq b$
- For any constraint $\sum_{j=1}^n a_{ij}x_j \geq b$ is replaced by $\sum_{j=1}^n -a_{ij}x_j \leq -b$.
- If a variable x_j has no sign restriction, it is replaced by $x_j = x_j^+ - x_j^-$ where $x_j^+, x_j^- \geq 0$.

Conversion to standard form: Example

- Consider the linear programming problem

$$\begin{array}{ll}\text{minimise} & -2x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

- convert min to max

$$\begin{array}{ll}\text{maxmise} & 2x_1 - 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

Conversion to standard form: Example

- Consider the linear programming problem

$$\begin{array}{ll}\text{minimise} & -2x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

- replace x_2 by $x_2 = x_2^+ - x_2^-$

$$\begin{array}{ll}\text{maximise} & 2x_1 - 3x_2^+ + 3x_2^- \\ \text{subject to} & x_1 + x_2^+ - x_2^- = 7 \\ & x_1 - 2x_2^+ + 2x_2^- \leq 4 \\ & x_1, x_2^+, x_2^- \geq 0\end{array}$$

Conversion to standard form: Example

- Consider the linear programming problem

$$\begin{array}{ll}\text{minimise} & -2x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

- replace = constraints by $\leq$ and $\geq$

$$\begin{array}{ll}\text{maxmise} & 2x_1 - 3x_2^+ + 3x_2^- \\ \text{subject to} & x_1 + x_2^+ - x_2^- \geq 7 \\ & x_1 + x_2^+ - x_2^- \leq 7 \\ & x_1 - 2x_2^+ + 2x_2^- \leq 4 \\ & x_1, x_2^+, x_2^- \geq 0\end{array}$$

Conversion to standard form: Example

- Consider the linear programming problem

$$\begin{array}{ll}\text{minimise} & -2x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

- Replace $\geq$ by $\leq$.

$$\begin{array}{ll}\text{maximise} & 2x_1 - 3x_2^+ + 3x_2^- \\ \text{subject to} & -x_1 - x_2^+ + x_2^- \leq -7 \\ & x_1 + x_2^+ - x_2^- \leq 7 \\ & x_1 - 2x_2^+ + 2x_2^- \leq 4 \\ & x_1, x_2^+, x_2^- \geq 0\end{array}$$

Conversion to standard form: Example

- Consider the linear programming problem

$$\begin{array}{ll}\text{minimise} & -2x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0\end{array}$$

- Rename x_2^+ and x_2^- as x_2 and x_3 .

$$\begin{array}{ll}\text{maximise} & 2x_1 - 3x_2 + 3x_3 \\ \text{subject to} & -x_1 - x_2 + x_3 \leq -7 \\ & x_1 + x_2 - x_3 \leq 7 \\ & x_1 - 2x_2 + 2x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

Conversion to slack form

- To solve the LP using the **simplex** method it is convenient to write it in **slack** form.
- This is done by converting the inequalities to equalities by introducing **slack** variables.
- For each constraint $\sum_{j=1}^n a_{ij}x_j \leq b_i$ introduce a slack variable s_i such that $s_i = b_i - \sum_{j=1}^n a_{ij}x_j$ with $s_i \geq 0$.
- As we will see later the slack variables will be mixed with the original variables during the iterations of the simplex method.
- Therefore we rename the slack variables s_i as x_{n+i} .

Conversion to slack form: Example

- As an example consider the linear programming problem in **standard form**

$$\begin{aligned} &\text{maximise } z = 3x_1 + x_2 + 2x_3 \\ &\text{subject to } x_1 + x_2 + 3x_3 \leq 30 \\ &\quad \quad \quad 2x_1 + 2x_2 + 5x_3 \leq 24 \\ &\quad \quad \quad 4x_1 + x_2 + 2x_3 \leq 36 \\ &\quad \quad \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

- We introduce 3 slack variables x_4, x_5, x_6

$$\begin{aligned} &\text{maximise } z = 3x_1 + x_2 + 2x_3 \\ &\text{subject to } x_1 + x_2 + 3x_3 + x_4 = 30 \\ &\quad \quad \quad 2x_1 + 2x_2 + 5x_3 + x_5 = 24 \\ &\quad \quad \quad 4x_1 + x_2 + 2x_3 + x_6 = 36 \\ &\quad \quad \quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

Basic and non-basic variables

- We rewrite the LP system in the form

$$z = 3x_1 + x_2 + 2x_3$$

$$x_4 = 30 - x_1 - x_2 - 3x_3$$

$$x_5 = 24 - 2x_1 - 2x_2 - 5x_3$$

$$x_6 = 36 - 4x_1 - x_2 - 2x_3$$

- The variables on the left side (x_4, x_5, x_6) are called **basic** variables and the variables on the right side (x_1, x_2, x_3) are called **non-basic** variables.
- By setting all the non-basic variables to 0 we obtain a **basic solution**.
- In this case $x_4 = 30, x_5 = 24, x_6 = 36$ and $z = 0$.
- A **basic solution** is said to be **basic feasible solution** if it satisfies all non-negative constraints.

Simplex Method

- The **simplex** method is an iterative algorithm for solving linear programming problems.
- Each iteration of the **simplex** rewrites the system to obtain a new **basic feasible solution** which usually has a higher value of z .
- This is done by selecting some variable x_e that has a positive coefficient in the objective function and increase it as much as possible without violating the constraints.
- The variable x_e becomes basic and some other basic variable x_l becomes non-basic.

Simplex method: Example

- Consider the previous problem again. The initial basic solution was
- $x_1 = x_2 = x_3 = 0, z = 0, x_4 = 30, x_5 = 24, x_6 = 36$
- We can increase the value of z by increasing the value of any of the non-basic variables. We choose to increase x_1 .
- How much can we increase x_1 without violating the constraints: $x_1 \leq 30, x_2 \leq 24/2, x_3 \leq 36/4$?
- Therefore x_1 will replace x_6 as a basic variable.

$$z = 3x_1 + x_2 + 2x_3$$

$$x_4 = 30 - x_1 - x_2 - 3x_3$$

$$x_5 = 24 - 2x_1 - 2x_2 - 5x_3$$

$$x_6 = 36 - 4x_1 - x_2 - 2x_3$$

Simplex method: Example

- Consider the previous problem again. The initial basic solution was
- $x_1 = x_2 = x_3 = 0, z = 0, x_4 = 30, x_5 = 24, x_6 = 36$
- We can increase the value of z by increasing the value of any of the non-basic variables. We choose to increase x_1 .
- How much can we increase x_1 without violating the constraints: $x_1 \leq 30, x_2 \leq 24/2, x_3 \leq 36/4$?
- Replace x_1 by its expression on the right hand side of the equations to obtain.

$$\begin{aligned}z &= 27 + \frac{x_2}{4} + \frac{x_3}{2} - \frac{3x_6}{4} \\x_1 &= 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \\x_4 &= 21 - \frac{3x_2}{4} - \frac{5x_3}{2} + \frac{x_6}{4} \\x_5 &= 6 - \frac{3x_2}{2} - 4x_3 + \frac{x_6}{2}\end{aligned}$$

Simplex method: Example

- Consider the previous problem again. The initial basic solution was
- $x_1 = x_2 = x_3 = 0, z = 0, x_4 = 30, x_5 = 24, x_6 = 36$
- We can increase the value of z by increasing the value of any of the non-basic variables. We choose to increase x_1 .
- How much can we increase x_1 without violating the constraints: $x_1 \leq 30, x_2 \leq 24/2, x_3 \leq 36/4$?
- We obtain a new **basic feasible solution**
 $x_1 = 9, x_2 = 0, x_3 = 0, x_4 = 21, x_5 = 6, x_6 = 0, z = 27.$

$$\begin{aligned}z &= 27 + \frac{x_2}{4} + \frac{x_3}{2} - \frac{3x_6}{4} \\x_1 &= 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \\x_4 &= 21 - \frac{3x_2}{4} - \frac{5x_3}{2} + \frac{x_6}{4} \\x_5 &= 6 - \frac{3x_2}{2} - 4x_3 + \frac{x_6}{2}\end{aligned}$$

example

- The value z can be increased by increasing the value of x_2 or x_3 . We choose x_3 , with the constraints $x_3 \leq 18$, $x_3 \leq 42/5$, $x_3 \leq 6/4$.
- x_3 will replace x_5 as a basic variable.

$$\begin{aligned}z &= \frac{111}{4} + \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16} \\x_1 &= \frac{33}{4} - \frac{x_2}{16} + \frac{x_5}{8} - \frac{5x_6}{16} \\x_3 &= \frac{3}{2} - \frac{3x_2}{8} - \frac{x_5}{4} + \frac{x_6}{8} \\x_4 &= \frac{69}{4} + \frac{3x_2}{16} + \frac{5x_5}{8} - \frac{x_6}{16}\end{aligned}$$

- We obtain a new **basic feasible solution** $z = 27.75$, $x_1 = 8.25$, $x_2 = 0$, $x_3 = 1.5$, $x_4 = 17.25$, $x_5 = 0$, $x_6 = 0$.

Simplex method: Example

- At this point z can increase only by increasing x_2 , with constraints $x_2 \leq 132$, $x_2 \leq 4$, $x_2 \leq \infty$ therefore x_3 is the leaving variable.

$$\begin{aligned}z &= \frac{111}{4} + \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16} \\x_1 &= \frac{33}{4} - \frac{x_2}{16} + \frac{x_5}{8} - \frac{5x_6}{16} \\x_3 &= \frac{3}{2} - \frac{3x_2}{8} - \frac{x_5}{4} + \frac{x_6}{8} \\x_4 &= \frac{69}{4} + \frac{3x_2}{16} + \frac{5x_5}{8} - \frac{x_6}{16}\end{aligned}$$

Simplex method: Example

$$\begin{aligned}z &= 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3} \\x_1 &= 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3} \\x_2 &= 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3} \\x_4 &= 18 - \frac{x_3}{2} + \frac{x_5}{2}\end{aligned}$$

- Since all coefficients in the objective function are negative, the current basic feasible solution is optimal. We are interested in the original variables and the objective function.
- $x_1 = 8$, $x_2 = 4$, $x_3 = 0$, $z = 28$.

Simplex Tableau

- The preceding operations can be done conveniently with the help of the **tableau** representation.

$$z - 3x_1 - x_2 - 2x_3 = 0$$

$$x_4 + x_1 + x_2 + 3x_3 = 30$$

$$x_5 + 2x_1 + 2x_2 + 5x_3 = 24$$

$$x_6 + 4x_1 + x_2 + 2x_3 = 36$$

	x_1	x_2	x_3	x_4	x_5	x_6	b
z	-3	-1	-2	0	0	0	0
x_4	1	1	3	1	0	0	30
x_5	2	2	5	0	1	0	24
x_6	4	1	2	0	0	1	36

Notice that the basic variables coefficients form the identity matrix.

Using tableau: Example

- The non-basic variables that increase the objective function are those with negative coefficients in the tableau representation.
- As before we choose x_1 to be the **entering** variable.
- The choice of the **leaving** variable corresponds to the row with the smallest ratio of b_1/a_{i1} . In this case it is x_6 .
- Then we perform a **pivot** operations: exchange x_1 and x_6 .

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	-3	-1	-2	0	0	0	0
x_4	1	1	3	1	0	0	30
x_5	2	2	5	0	1	0	24
x_6	4	1	2	0	0	1	36

- A **pivot** operation exchanges the **entering** variable with the **leaving** variable.
- In the tableau representation this is done by performing row operations to obtain a new (equivalent) tableau.
- The new tableau will have all coefficients in the entering variable column equal to zero except for the pivot element is equal to 1

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	-3	-1	-2	0	0	0	0
R_1	x_4	1	1	3	1	0	0	30
R_2	x_5	2	2	5	0	1	0	24
R_3	x_6	4	1	2	0	0	1	36

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
$(R_0 + 3R_3/4)z$	0	-1/4	-1/2	0	0	3/4	27
x_4	1	1	3	1	0	0	30
x_5	2	2	5	0	1	0	24
$x_6 \leftrightarrow x_1$	4	1	2	0	0	1	36

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	-3	-1	-2	0	0	0	0
R_1	x_4	1	1	3	1	0	0	30
R_2	x_5	2	2	5	0	1	0	24
R_3	x_6	4	1	2	0	0	1	36

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	-1/4	-1/2	0	0	3/4	27
$(R_1 - R_3/4)x_4$	0	3/4	5/2	1	0	-1/4	21
x_5	2	2	5	0	1	0	24
$x_6 \leftrightarrow x_1$	4	1	2	0	0	1	36

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	-3	-1	-2	0	0	0	0
R_1	x_4	1	1	3	1	0	0	30
R_2	x_5	2	2	5	0	1	0	24
R_3	x_6	4	1	2	0	0	1	36

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	-1/4	-1/2	0	0	3/4	27
x_4	0	3/4	5/2	1	0	-1/4	21
$(R_2 - R_3/2)x_5$	0	3/2	4	0	1	-1/2	6
$x_6 \leftrightarrow x_1$	4	1	2	0	0	1	36

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	-3	-1	-2	0	0	0	0
R_1	x_4	1	1	3	1	0	0	30
R_2	x_5	2	2	5	0	1	0	24
R_3	x_6	4	1	2	0	0	1	36

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
	z	0	-1/4	-1/2	0	0	3/4	27
	x_4	0	3/4	5/2	1	0	-1/4	21
	x_5	0	3/2	4	0	1	-1/2	6
$(R_3/4)$	x_1	1	1/4	1/2	0	0	1/4	9

Using tableau: Example

- The objective function can be increased further by increasing x_2 or x_3 . We choose x_3 as the entering variable
- The constraints are $x_3 \leq 2 \times 21/5$, $x_3 \leq 6/4$, $x_3 \leq 2 \times 9$ therefore x_5 is the leaving variable.

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	$-1/4$	$-1/2$	0	0	$3/4$	27
x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
x_5	0	$3/2$	4	0	1	$-1/2$	6
x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/4$	$-1/2$	0	0	$3/4$	27
R_1	x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
R_2	x_5	0	$3/2$	4	0	1	$-1/2$	6
R_3	x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
$(R_0 + R_2/8)z$		0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
	x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
	$x_5 \leftrightarrow x_3$	0	$3/2$	4	0	1	$-1/2$	6
	x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/4$	$-1/2$	0	0	$3/4$	27
R_1	x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
R_2	x_5	0	$3/2$	4	0	1	$-1/2$	6
R_3	x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
$(R_1 - 5R_2/8)x_4$	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
$x_5 \leftrightarrow x_3$	0	$3/2$	4	0	1	$-1/2$	6
x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/4$	$-1/2$	0	0	$3/4$	27
R_1	x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
R_2	x_5	0	$3/2$	4	0	1	$-1/2$	6
R_3	x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
$x_5 \leftrightarrow x_3$	0	$3/2$	4	0	1	$-1/2$	6
$(R_3 - R_2/8)x_1$	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/4$	$-1/2$	0	0	$3/4$	27
R_1	x_4	0	$3/4$	$5/2$	1	0	$-1/4$	21
R_2	x_5	0	$3/2$	4	0	1	$-1/2$	6
R_3	x_1	1	$1/4$	$1/2$	0	0	$1/4$	9

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
	z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
$(R_2/4)$	x_3	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

Using tableau: Example

- At this point the only variable that can increase the objective function is x_2 so x_2 is the entering variable.
- Using the constraints $x_2 \leq 4$, $x_2 \leq 132$, and $x_2 < \infty$ we find that x_3 is the leaving variable.

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	-1/16	0	0	1/8	11/16	111/4
x_4	0	-3/16	0	1	-5/8	1/16	69/4
x_3	0	3/8	1	0	1/4	-1/8	3/2
x_1	1	1/16	0	0	-1/8	5/16	33/4

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
R_1	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
R_2	x_3	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
R_3	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
$(R_0 + R_2/6)$	z	0	0	$1/6$	0	$1/6$	$2/3$	28
	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
	$x_3 \leftrightarrow x_2$	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
R_1	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
R_2	x_3	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
R_3	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
	z	0	0	$1/6$	0	$1/6$	$2/3$	28
$(R_1 + R_2/2)$	x_4	0	0	$1/2$	1	$-1/2$	0	18
	$x_3 \leftrightarrow x_2$	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
R_1	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
R_2	x_3	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
R_3	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
	z	0	0	$1/6$	0	$1/6$	$2/3$	28
	x_4	0	0	$1/2$	1	$-1/2$	0	18
	$x_3 \leftrightarrow x_2$	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
$(R_3 - R_2/6)$	x_1	1	0	$-1/6$	0	$-1/6$	$1/3$	8

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	$-1/16$	0	0	$1/8$	$11/16$	$111/4$
R_1	x_4	0	$-3/16$	0	1	$-5/8$	$1/16$	$69/4$
R_2	x_3	0	$3/8$	1	0	$1/4$	$-1/8$	$3/2$
R_3	x_1	1	$1/16$	0	0	$-1/8$	$5/16$	$33/4$

$(8R_2/3)$

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	0	$1/6$	0	$1/6$	$2/3$	28
x_4	0	0	$1/2$	1	$-1/2$	0	18
x_2	0	1	$8/3$	0	$2/3$	$-1/3$	4
x_1	1	0	$-1/6$	0	$-1/6$	$1/3$	8

Using tableau: Example

		x_1	x_2	x_3	x_4	x_5	x_6	RHS
R_0	z	0	-1/16	0	0	1/8	11/16	111/4
R_1	x_4	0	-3/16	0	1	-5/8	1/16	69/4
R_2	x_3	0	3/8	1	0	1/4	-1/8	3/2
R_3	x_1	1	1/16	0	0	-1/8	5/16	33/4

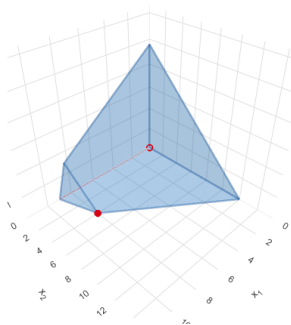
		x_1	x_2	x_3	x_4	x_5	x_6	RHS
	z	0	0	1/6	0	1/6	2/3	28
	x_4	0	0	1/2	1	-1/2	0	18
$(8R_2/3)$	x_2	0	1	8/3	0	2/3	-1/3	4
	x_1	1	0	-1/6	0	-1/6	1/3	8

All coefficients of z are >0 . Optimal solution reached

$$z = 28, x_1 = 8, x_2 = 4, x_3 = 0$$

Geometric interpretation

- The simplex algorithm can be interpreted geometrically as moving from one vertex of the feasible region to another as shown below.
- Note that in this example x_2 was chosen as entering variable before x_3 for a total of 2 pivot operations instead of 3
- $(0, 0, 0) \rightarrow (9, 0, 0) \rightarrow (8, 4, 0)$



Simplex algorithm

- If $c_j \geq 0$ for all $j = 1 \dots n$ then the optimal solution is $z = b_0$
- Otherwise find a column with $c_s < 0$, x_s is the entering variable.
- If $a_{is} \leq 0$ for all i then the problem is unbounded: there is no maximum
- (Min. ratio test) Pivot on a_{rs} , x_r is the leaving variable, where

$$\frac{b_r}{a_{rs}} = \min_{i: a_{is} > 0} \frac{b_i}{a_{is}}$$

	x_1	$\dots$	x_s	$\dots$	x_n	
z	c_1	$\dots$	c_s	$\dots$	c_n	b_0
	a_{11}	$\dots$	a_{1s}	$\dots$	a_{1n}	b_1
	$\vdots$		$\vdots$		$\vdots$	$\vdots$
	a_{r1}	$\dots$	a_{rs}	$\dots$	a_{rn}	b_r
	$\vdots$		$\vdots$		$\vdots$	$\vdots$
	a_{m1}	$\dots$	a_{ms}	$\dots$	a_{mn}	b_m

Unbounded feasible region

- Sometimes the feasible region is unbounded. In that case the objective function can be increased indefinitely, i.e. there is no maximum.
- As an example consider the LP problem:

$$\begin{aligned} &\text{maximize } 2x_1 + x_2 \\ &\text{subject to } x_1 - x_2 \leq 10 \\ &\quad \quad \quad 2x_1 - x_2 \leq 40 \\ &\quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$$

- Convert it to slack form :

$$\begin{aligned} &\text{maximize } 2x_1 + x_2 \\ &\text{subject to } x_1 - x_2 + x_3 = 10 \\ &\quad \quad \quad 2x_1 - x_2 + x_4 = 40 \\ &\quad \quad \quad x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Unbounded feasible region

- Choose x_1 as the entering variable. The minimum ratio test gives $b_1/a_{11} = 10/1 = 10$ and $b_2/a_{21} = 40/2 = 20$. Therefore x_3 is the leaving variable.

		x_1	x_2	x_3	x_4	RHS
R_0	z	-2	-1	0	0	0
R_1	x_3	1	-1	1	0	10
R_2	x_4	2	-1	0	1	40

	x_1	x_2	x_3	x_4	RHS
$(R_0 + 2R_1) \ z$	0	-3	2	0	20
$x_3 \leftrightarrow x_1$	1	-1	1	0	10
x_4	2	-1	0	1	40

Unbounded feasible region

		x_1	x_2	x_3	x_4	RHS
R_0	z	-2	-1	0	0	0
R_1	x_3	1	-1	1	0	10
R_2	x_4	2	-1	0	1	40

		x_1	x_2	x_3	x_4	RHS
	z	0	-3	2	0	20
	x_1	1	-1	1	0	10
$(R_2 - 2R_1)$	x_4	0	1	-2	1	20

Unbounded feasible region

- x_2 is the only choice to increase the objective function. The minimum ratio test gives $b_2/a_{22} = 20/1 = 20$ (only positive ratios are considered). Therefore x_4 is the leaving variable.

		x_1	x_2	x_3	x_4	RHS
R_0	z	0	-3	2	0	20
R_1	x_1	1	-1	1	0	10
R_2	x_4	0	1	-2	1	20

	x_1	x_2	x_3	x_4	RHS
$(R_0 + 3R_2) \ z$	0	0	-4	3	80
x_1	1	-1	1	0	10
$x_4 \leftrightarrow x_2$	0	1	-2	1	20

Unbounded feasible region

		x_1	x_2	x_3	x_4	RHS
R_0	z	0	-3	2	0	20
R_1	x_1	1	-1	1	0	10
R_2	x_4	0	1	-2	1	20

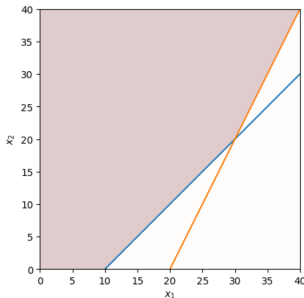
		x_1	x_2	x_3	x_4	RHS
	z	0	0	-4	3	80
	x_1	1	0	-1	1	30
$(R_1 + R_2)$	x_2	0	1	-2	1	20

Entering variable x_3 but

$\frac{b_i}{a_{i3}} \leq 0$ for all i then the problem is unbounded

Unbounded feasible region

- The unboundedness of the solution can be seen visually by plotting the feasible region and constraints
- The objective function has **no maximum**. The simplex steps go from $(0,0)$ to $(10,0)$ to $(30,20)$. After that the objective function increases without bound.



Initial basic feasible solution

- Not all LP problems have a feasible solution.
- Moreover, even when a problem has a feasible solution the correctness of the simplex method depends on the **initial basic solution being feasible**.
- Next we describe a procedure to determine if an LP problem is feasible.
- In addition, if the problem is feasible the procedure returns a slack form that has a feasible initial basic solution.

Example

Consider the following linear program:

$$\begin{aligned} &\text{maximize} && 2x_1 - x_2 \\ &\text{subject to} && 2x_1 - x_2 \leq 2 \\ & && x_1 - 5x_2 \leq -4 \\ & && x_1, x_2 \geq 0 \end{aligned}$$

And in slack form:

$$\begin{aligned} &\text{maximize} && 2x_1 - x_2 \\ &\text{subject to} && 2x_1 - x_2 + x_3 = 2 \\ & && x_1 - 5x_2 + x_4 = -4 \\ & && x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Example

- The initial basic solution is feasible if $b_i \geq 0$ for all i .
- In this case $b_1 = 2$ and $b_2 = -4$, not both ≥ 0 .
- We can convert the problem to a form that has a feasible initial basic solution.
- When we don't have a feasible initial basic solution we use the so called **two-phase simplex method**.
 - 1 Phase 1: Find a feasible solution by minimizing an auxiliary function.
 - 2 Phase 2: Use the feasible solution from phase 1 to solve the original problem.

Phase 1: auxiliary problem

- Introduce a new variable x_0 and consider the following auxiliary problem:

$$\begin{array}{ll}\text{maximize} & -x_0 \\ \text{subject to} & \sum_{j=1}^n a_{ij}x_j - x_0 \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 0, 1, \dots, n\end{array}$$

- The original problem is feasible iff the auxiliary has a solution with $x_0 = 0$.
- Note that when solving the auxiliary problem the **first** pivot operation always chooses x_0 as the entering variable.

Auxiliary problem: Example

Continuing with our previous example, the auxiliary problem is:

$$\begin{aligned} &\text{maximize} && -x_0 \\ &\text{subject to} && 2x_1 - x_2 - x_0 \leq 2 \\ & && x_1 - 5x_2 - x_0 \leq -4 \\ & && x_0, x_1, x_2 \geq 0 \end{aligned}$$

And in slack form:

$$\begin{aligned} &\text{maximize} && -x_0 \\ &\text{subject to} && 2x_1 - x_2 - x_0 + x_3 = 2 \\ & && x_1 - 5x_2 - x_0 + x_4 = -4 \\ & && x_0, x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Auxiliary problem: Example

- In tableau form:

	x_1	x_2	x_0	x_3	x_4	RHS
z	0	0	1	0	0	0
x_3	2	-1	-1	1	0	2
x_4	1	-5	-1	0	1	-4

- The auxiliary problem **does not** have an initial feasible solution.
- We get an initial BFS by pivoting on x_0 and the basic variable corresponding to the "most" negative b_i is the leaving variable. In this case x_4

Auxiliary problem: Example

After pivoting on x_0 we have a BFS and we proceed with the simplex method.

	x_1	x_2	x_0	x_3	x_4	RHS
z	1	-5	0	0	1	-4
x_3	1	4	0	1	-1	6
x_0	-1	5	1	0	-1	4

Pivot $x_2 \leftrightarrow x_0$:

	x_1	x_2	x_0	x_3	x_4	RHS
z	0	0	1	0	0	0
x_3	$9/5$	0	$-4/5$	1	$-1/5$	$14/5$
x_2	$-1/5$	1	$1/5$	0	$-1/5$	$4/5$

Since $x_0 = 0, z = 0$ the original problem is feasible.

Phase 2: Original problem

- We have obtained a initial BFS and we proceed with simplex for the original problem.
- This is done by dropping x_0 and recovering the original objective function: $2x_1 - x_2$.
- The objective function is written in terms of the non-basic variables (x_1, x_4) , recall that $x_1 - 5x_2 + x_4 = -4$.

Starting tableau for phase 2:

	x_1	x_2	x_3	x_4	RHS
z	$-9/5$	0	0	$1/5$	$-4/5$
x_3	$9/5$	0	1	$-1/5$	$14/5$
x_2	$-1/5$	1	0	$-1/5$	$4/5$

Phase 2: Example

	x_1	x_2	x_3	x_4	RHS
z	-9/5	0	0	1/5	-4/5
x_3	9/5	0	1	-1/5	14/5
x_2	-1/5	1	0	-1/5	4/5

Pivot $x_1 \leftrightarrow x_3$:

	x_1	x_2	x_3	x_4	RHS
z	0	0	1	0	2
x_1	1	0	5/9	-1/9	14/9
x_2	0	1	1/9	-2/9	10/9

Therefore the optimal solution is $x_1 = 14/9$, $x_2 = 10/9$, $z = 2$.

- An LP system is degenerate if one of the b 's is zero.
- A pivot is said to be degenerate when the leaving variable has zero ratio.
- Sometimes (not always) degenerate pivots cause the simplex method to cycle.
- Let T_i be a tableau at iteration i . If $T_i = T_j$ for some $j < i$ then the simplex method is cycling.
- This means that the simplex will never terminate.

Degeneracy: Example

Consider the following LP problem:

$$\begin{array}{ll}\text{maximise} & x_1 - 2x_2 - 2x_4 \\ \text{subject to} & 0.5x_1 - 3.5x_2 - 2x_3 + 4x_4 \leq 0 \\ & 0.5x_1 - x_2 - 0.5x_3 + 0.5x_4 \leq 0 \\ & x_1 \leq 1 \\ & x_1, x_2, x_3, x_4 \geq 0\end{array}$$

Corresponding tableau:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-1	2	0	2	0	0	0	0
x_5	1/2	-7/2	-2	4	1	0	0	0
x_6	1/2	-1	-1/2	1/2	0	1	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Pivot $x_1 \leftrightarrow x_5$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-1	2	0	2	0	0	0	0
x_5	0.5	-3.5	-2	4	1	0	0	0
x_6	0.5	-1	-0.5	0.5	0	1	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Result

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	-5	-4	10	2	0	0	0
x_1	1	-7	-4	8	2	0	0	0
x_6	0	$5/2$	$3/2$	$-7/2$	-1	1	0	0
x_7	0	7	4	-8	-2	0	1	1

Simplex cycle

Pivot $x_2 \leftrightarrow x_6$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	-5	-4	10	2	0	0	0
x_1	1	-7	-4	8	2	0	0	0
x_6	0	$5/2$	$3/2$	$-7/2$	-1	1	0	0
x_7	0	7	4	-8	-2	0	1	1

Simplex cycle

Result

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	-1	3	0	2	0	0
x_1	1	0	1/5	-9/5	-4/5	14/5	0	0
x_2	0	1	3/5	-7/5	-2/5	2/5	0	0
x_7	0	0	-1/5	9/5	4/5	-14/5	1	1

Simplex cycle

Pivot $x_3 \leftrightarrow x_1$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	-1	3	0	2	0	0
x_1	1	0	1/5	-9/5	-4/5	14/5	0	0
x_2	0	1	3/5	-7/5	-2/5	2/5	0	0
x_7	0	0	-1/5	9/5	4/5	-14/5	1	1

Simplex cycle

Result

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	5	0	0	-6	-4	16	0	0
x_3	5	0	1	-9	-4	14	0	0
x_2	-3	1	0	4	2	-8	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Pivot $x_4 \leftrightarrow x_2$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	5	0	0	-6	-4	16	0	0
x_3	5	0	1	-9	-4	14	0	0
x_2	-3	1	0	4	2	-8	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Result

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	$1/2$	$3/2$	0	0	-1	4	0	0
x_3	$-7/4$	$9/4$	1	0	$1/2$	-4	0	0
x_4	$-3/4$	$1/4$	0	1	$1/2$	-2	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Pivot $x_5 \leftrightarrow x_3$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	$1/2$	$3/2$	0	0	-1	4	0	0
x_3	$-7/4$	$9/4$	1	0	$1/2$	-4	0	0
x_4	$-3/4$	$1/4$	0	1	$1/2$	-2	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Result

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-3	6	2	0	0	-4	0	0
x_5	$-7/2$	$9/2$	2	0	1	-8	0	0
x_4	1	-2	-1	1	0	2	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

Pivot $x_6 \leftrightarrow x_4$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-3	6	2	0	0	-4	0	0
x_5	$-7/2$	$9/2$	2	0	1	-8	0	0
x_4	1	-2	-1	1	0	2	0	0
x_7	1	0	0	0	0	0	1	1

Simplex cycle

The result is the same tableau we started with. Simplex will cycle indefinitely

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-1	2	0	2	0	0	0	0
x_5	1/2	7/2	-2	4	1	0	0	0
x_6	1/2	-1	-1/2	1/2	0	1	0	0
x_7	1	0	0	0	0	0	1	1

Bland pivot rule

- Simplex is guaranteed to terminate if we use the **Bland pivot rule**.
- Among all possible entering variables, choose the one with the smallest index.
- In fact in the previous example, we followed the Bland rule except in the last step where we chose x_6 as the entering variable instead of x_1

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-3	6	2	0	0	-4	0	0
x_5	$-7/2$	$9/2$	2	0	1	-8	0	0
x_4	1	-2	-1	1	0	2	0	0
x_7	1	0	0	0	0	0	1	1

Bland rule

Pivot $x_1 \leftrightarrow x_4$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-3	6	2	0	0	-4	0	0
x_5	-7/2	9/2	2	0	1	-8	0	0
x_4	1	-2	-1	1	0	2	0	0
x_7	1	0	0	0	0	0	1	1

Bland rule

Result:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	-1	3	0	2	0	0
x_5	0	$-5/2$	$-3/2$	$7/2$	1	-1	0	0
x_1	1	-2	-1	1	0	2	0	0
x_7	0	2	1	-1	0	-2	1	1

Pivot $x_3 \leftrightarrow x_7$:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	-1	3	0	2	0	0
x_5	0	-5/2	-3/2	7/2	1	-1	0	0
x_1	1	-2	-1	1	0	2	0	0
x_7	0	2	1	-1	0	-2	1	1

Result:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	2	0	2	0	0	1	1
x_5	0	1/2	0	2	1	-4	3/2	3/2
x_1	1	0	0	0	0	0	1	1
x_3	0	2	1	-1	0	-2	1	1

Simplex terminated with $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 0$ and $z = 1$

Duality

- Given an LP problem (LP1) in standard form:

$$\begin{aligned} & \text{maximize} && \sum_{j=1}^n c_j x_j \\ & \text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i && \text{for } i = 1, 2, \dots, m \quad (1) \\ & && x_j \geq 0 && \text{for } j = 1, 2, \dots, n \end{aligned}$$

- We associate the dual (LP2):

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^m b_i y_i \\ & \text{subject to} && \sum_{i=1}^m a_{ij} y_i \geq c_j && \text{for } j = 1, 2, \dots, n \quad (2) \\ & && y_i \geq 0 && \text{for } i = 1, 2, \dots, m \end{aligned}$$

- **Weak duality theorem:**

- If LP1 is unbounded then LP2 is infeasible (and vice versa).
- If LP1 is infeasible then LP2 is either unbounded or infeasible (and vice versa).
- If both LP1 and LP2 are bounded and feasible then:

$$\text{opt}(LP1) \leq \text{opt}(LP2)$$

- **Strong duality theorem:**

- if either LP1 or LP2 is bounded and feasible then so is the other and:

$$\text{opt}(LP1) = \text{opt}(LP2)$$

Fundamental theorem of linear programming

- The fundamental theorem of linear programming states that the extrema of a linear function over a convex polygonal region occurs at the region's vertices.
- Furthermore, if an extreme occurs at two adjacent vertices then it occurs everywhere on the line segment connecting them (i.e., infinite number of solutions).
- If the feasible region is unbounded the problem might not have a solution.
- In general the problem has multiple solutions when the objective function is parallel to one of the constraints

Convergence and Complexity of Simplex method

- The simple pivoting rule used in this lecture converges to the optimal solution in almost all cases (except possibly for degenerate cases)
- With the **appropriate** choice of pivot rules (e.g. Bland rule) Simplex is guaranteed to converge.
- In the worst case the complexity of Simplex is exponential in the number of variables.
- While one can construct problems where Simplex takes exponential time, in practice Simplex is quite efficient.
- Check out [▶ this link](#) for further discussion

All the material is taken from your textbook.

- Can solve LP using `scipy.optimize.linprog` function in `scipy` Python package
- [linprog documentation](#)
- There is also the interactive package [GILP](#)
- Follow [this link](#) for an example of both. Click on "Open in Colab" to run it on Google Colab (need to log in to Google account)
- A nice tool to practice simplex can be found [here](#)
- A simple LP solver using `linprog` Python package can be found [here](#)